

Jishnu Nambissan Thekkeppattu (he/him)

CONTACT INFORMATION	CSIRO Space and Astronomy, 26 Pembroke Road, Marsfield, NSW 2122, Australia. jishnunambissan.thekkeppattu@csiro.au	
RESEARCH INTERESTS	Radio astronomy, radio interferometry, instrumentation and techniques. Phased arrays, calibration and array signal processing, wide-field polarimetry. High speed DSP, reconfigurable computing, FPGAs. Experimental and observational 21-cm cosmology, precision radiometry, data analysis for 21-cm experiments, radio-sky modelling at low-frequencies. All-sky spherical harmonic transit interferometry (<i>m</i> -mode imaging). Numerical linear algebra, ill-posed inverse problems, iterative methods. Ionospheric scintillation.	
PROFESSIONAL EXPERIENCE	Aug 2025 - Present Aug 2022 - Jul 2025 Nov 2016 - Jul 2022 Jan 2016 - Oct 2016	CERC Fellow, CSIRO Space and Astronomy, Australia Research Associate, Curtin University, Australia Research Scientist/Research Associate, Raman Research Institute, India Assistant System Engineer, Tata Consultancy Services, India
EDUCATION	Curtin University, Australia Doctor of Philosophy – Physics (2019–2022) • Thesis title : Towards detection of redshifted 21-cm signal from Cosmic Dawn and Epoch of Reionization. • Thesis supervisors : Dr. Cathryn M. Trott, Dr. Benjamin McKinley (Curtin Institute of Radio Astronomy, Bentley, WA) Kannur University, India Bachelor of Technology in Electronics and Communication Engineering (2011–2015) • Aggregate percentage : 81.2% • Undergraduate project : A VHF board for Software Defined Radio (SDR) applications. • Project advisor: Dr. V. Vinodkumar (Govt. College of Engineering, Kannur)	

RESEARCH
EXPERIENCE

- 2025-Present Development of an ultrafast radio camera using FPGAs
Supervisor(s): Dr. Nithyanandan Thyagarajan, Dr. David Humphrey
CSIRO Space and Astronomy, Marsfield, NSW.
- 2022-2025 SKA-Low bridging and Early Production Array activities &
All-sky mapping with spherical harmonic transit interferometry
Supervisor(s): Dr. Randall Wayth, Dr. David Davidson
Curtin Institute of Radio Astronomy, Bentley, WA.
- 2019-2022 Towards detection of redshifted 21-cm signal from Cosmic Dawn
and Epoch of Reionization.
Supervisor(s): Dr. Cathryn Trott, Dr. Benjamin McKinley,
Curtin Institute of Radio Astronomy, Bentley, WA.
- 2019-2021 Principal investigator of PRATUSH: Probing ReionizATIOn of the
Universe using Signal from Hydrogen.
(Funded for pre-project studies by the ISRO)
Co-PIs : Dr. Saurabh Singh, Dr. Mayuri S. Rao
Raman Research Institute, Bangalore, India.
- 2016-2021 On the detection of global 21-cm signal from Cosmic Dawn and
Epoch of Reionization.
Advisors: Dr. Ravi Subrahmanyam, Dr. N. Udaya Shankar,
Raman Research Institute, Bangalore, India.
- 2019 June- Comparing LOFAR Epoch of Reionization target fields using 21-cm
August power spectra.
Advisor : Dr. André Offringa,
ASTRON, Dwingeloo, The Netherlands.
- 2014-2015 Reconfigurable radios using Field Programmable Gate Arrays.
Advisor: Dr. V. Vinodkumar,
Govt. College of Engineering, Kannur, India
- 2014-2015 A VHF board for Software Defined Radio (SDR) applications.
Advisor: Dr. V. Vinodkumar,
Govt. College of Engineering, Kannur, India
- 2013-2014 Multiplierless filter realisation using Canonic Signed Digits.
Advisor: Dr. Manju Manuel,
Govt. College of Engineering, Kannur, India

SELECTED TALKS	2026 Aug	<i>Monitoring equatorial ionospheric scintillation using low-cost software defined radios - some results from the Indian sector</i> - URSI GASS 2026, held at Krakow, Poland (YSA talk)
	2026 Aug	<i>Ultrafast real-time imaging with large-N aperture arrays using FPGA accelerators</i> - URSI GASS 2026, held at Krakow, Poland
	2025 Dec	<i>Ultra-fast real-time imaging with large aperture arrays</i> - SALF-XI, held at Orleans, France (remote talk)
	2025 Aug	<i>All-sky m-mode map making at 159 MHz with EDA2 and SE607HBA LOFAR station</i> - URSI AP-RASC 2025, held at Sydney, Australia
	2025 Aug	<i>Broadband self-holography of EDA2 - an SKA-Low prototype station</i> - URSI AP-RASC 2025, held at Sydney, Australia
	2025 Jun	<i>All-sky map making with spherical harmonics in the SKA-Low era</i> - SKAO General Science Meeting, A new era in astrophysics: Preparing for early science with the SKAO, held at Goerlitz, Germany
	2020 Jan	<i>Updates from the SARAS experiment</i> - International School And Conference On Observing The First Billion Years Of The Universe Using The Next Generation Telescopes, held at IIT Indore, India (remote talk)
HONORS AND AWARDS	2026	Young Scientist Award, URSI-GASS-2026
	2022	Joint winner of the ICRAR Ken and Julie Michael Prize
	2014	Best student paper award in embedded systems and VLSI category at National Conference on Systems Energy and Environment - 14 held at Govt College of Engineering, Kannur
	2010	Indian Space Research Organisation (ISRO) sponsored student delegate, 97 th Indian Science Congress held at Kerala University.
TEACHING	2025	COMT4003 - Mobile Radio Communications; Unit Co-ordinator & Lecturer (Curtin University)
	2024	COMT6005 - Satellite Communications; Lecturer (Curtin University)
	2024	COMT4003 - Mobile Radio Communications; Lecturer (Curtin University)
	2023	COMT6005 - Satellite Communications; Lecturer (Curtin University)
	2020	ELEN3002/6017 - Engineering Electromagnetic Labs (Curtin University)
MEMBERSHIPS	2023-Present	Junior member - International Astronomical Union (IAU)
	2024-2025	Member - Institute of Electrical and Electronics Engineers (IEEE)
	2013-2014	Student member - IEEE
OBSERVING AND FIELD EXPERIENCE	Deployment, operation and maintenance of a prototype two-element broadband interferometer SITARA at Inyarrimanha Ilgari Bundara (the Murchison Radio-astronomy Observatory) (2021-2022).	
	Measurement of global sky brightness temperature at low frequencies deploying SARAS at remote sites at Timbaktu Collective, India (2017-18), Indian Astronomical Obser-	

vatory, Ladakh, India (2017-18) and Mavathur lake, India (2019).

Electromagnetic tests and measurements of performance and characteristics of the SARAS antennas and receivers at the Gauribidanur Field Station, India (2017-19) and Mavathur lake, India (2019).

HARDWARE AND LABORATORY EXPERIENCE	Expertise in electronics and RF/microwave systems; using vector network analysers (Keysight FieldFox, PNA-X etc.) and their calibration, spectrum analysers, system level RF design, development and testing. Hands-on experience in handling fibre-optics, optical power measurements and RFoF systems. Experienced in performing measurements in an electromagnetic compatibility (EMC) chamber. Expertise in basic laboratory/workshop tasks such as soldering and miscellaneous mechanical works. FPGA systems (Zynq, Alveo v80).	
SOFTWARE TOOLS AND PACKAGES	Programming Languages:	python, C, C++, bash scripting
	HDL:	VHDL
	Software packages:	miriad , MATLAB
	Operating systems:	Linux, Mac OS
	Some experience in working with a few other languages (x86 and 8051 assembly, embedded C for AVR microcontrollers, arduino, Verilog, Visual C# etc.), EM simulators (NEC2, CST) and HPC systems	
OUTREACH	2025–Present	CSIRO S&A Marsfield colloquium organiser.
	2020–2025	DIY radio telescope programme. In collaboration with the Curtin ASTRO-3D node, built a low-cost hobby radio telescope operating at 1420 MHz for outreach purposes.
	2015	Institute of Electrical and Electronics Engineers - Teachers In Service programme (IEEE-TISP) volunteer. Conducted science and engineering outreach programmes for high school students.
VOLUNTEERING AND HOBBIES	2021-Present	Australian Red Cross Lifeblood blood, platelets & plasma donor
	2014-Present	Amateur radio operator, holding Indian callsign VU3VWB and Australian callsign VK6JN